

GENERATIVE AI · DIGITAL TRUST & FRAUD INTELLIGENCE

Verify Before You Trust.

An explainable Generative-AI platform that investigates digital deception — phishing, scams, impersonation and fraud — before it causes harm.

 github.com/Showkat1/trustiva-ai

 trustiva-ai.streamlit.app



THE PROBLEM

Digital deception is winning

Every day, people receive messages, links, job offers and investment "opportunities" that are engineered to deceive. Across SMS, WhatsApp, email and social media, it is harder than ever to tell what is real.



The gap

Existing tools just stamp a message "spam" or "fraud". They don't verify the claim, explain their reasoning, or tell the victim what to do next — so people are left guessing.



Phishing messages

Fake bank / OTP / account alerts



Fake offers

Job, prize & investment schemes



Malicious links

Look-alike & shortened URLs



Impersonation

Banks, govt bodies & brands

THE SOLUTION

Trustiva investigates — it doesn't just label

From

“this looks like spam”



to a defensible, evidence-based investigation with a clear next step



Verifies suspicious claims

Reasons over evidence instead of matching keywords.



Correlates risk signals

Connects URLs, requests, urgency & impersonation.



Explainable trust score

A transparent 0–100 score with a “why” for every case.

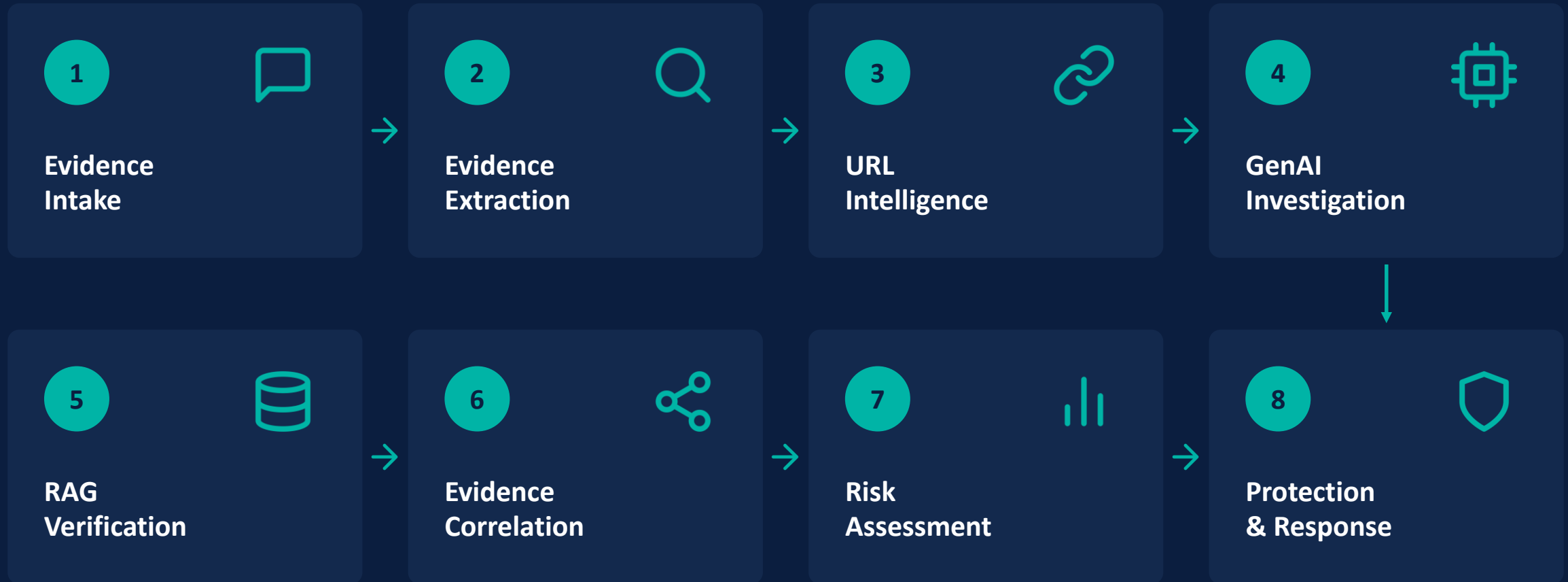


Prevention & recovery

Personalised, risk-tiered protection actions.

HOW IT WORKS

An 8-step investigation pipeline



Every stage is transparent and inspectable in the app — extracted evidence, retrieved sources, correlation and the final decision are all shown to the user.

CAPABILITIES

What's inside



Multimodal intake

Analyse a message, a URL, a screenshot (OCR) or a PDF / TXT document.



GenAI investigation

Structured, evidence-grounded reasoning returned as strict JSON.



Trusted-source RAG

FAISS semantic search over a curated cyber-fraud knowledge base.



URL intelligence

Deterministic checks: HTTPS, punycode, shorteners, brand mismatch.



Evidence correlation

Links individual signals into one explainable investigation picture.



Score + actions

A 0–100 risk score plus tiered prevention & recovery guidance.

UNDER THE HOOD

Technology stack



Interface

- Streamlit web app
- Custom CSS UI



Generative AI

- OpenRouter LLM
- OpenAI-compatible client
- Temp 0 · strict JSON



Retrieval (RAG)

- Sentence-Transformers
- all-MiniLM-L6-v2
- FAISS vector index



Vision & documents

- Tesseract OCR (pytesseract)
- PyMuPDF PDF reader



Analysis engine

- Regex evidence extraction
- Pydantic data models
- Explainable risk engine



Quality & deploy

- pytest + evaluation suite
- Streamlit Community Cloud

High-level architecture



DIFFERENTIATOR

Explainable — not a black box

How the score is built



Weighted signals

Credential requests, payment demands, suspicious URLs and urgency each carry a calibrated weight.



Correlation bonuses

Dangerous combinations (e.g. credentials + suspicious link) raise the score beyond their parts.



Anti-double-counting

Contextual labels are weighted low, so one scam narrative can't inflate its own score.



Critical gating

Reaching CRITICAL requires genuinely independent, corroborating evidence.

Transparent risk bands

CRITICAL

80–100

HIGH

50–79

MEDIUM

25–49

LOW

0–24

Every result ships with a plain-language “Why this score?” and the exact evidence behind it.

SEE IT LIVE

Live demo



INCOMING MESSAGE

"Your bank account has been suspended. Click <https://secure-bank-login.com> immediately and enter your OTP and password to restore access."

What Trustiva shows

- ✓ Structured evidence: URL, credential & OTP request
- ✓ URL intelligence flags a look-alike login domain
- ✓ GenAI investigation classifies the threat category
- ✓ RAG cites trusted phishing-awareness sources
- ✓ Signals correlate → CRITICAL risk score
- ✓ Clear recovery actions: don't click, verify, report



Try it: trustiva-ai.streamlit.app

Responsible by design



Safety guardrails

- ✓ Evidence is treated as observation, never proof
- ✓ Never makes a definitive criminal accusation
- ✓ An unknown sender is not automatically fraud
- ✓ Human stays in the loop — guidance, not verdicts



Built for Pakistan

- ✓ PKR amounts & local phone-number formats
- ✓ Aware of SBP, FBR, NADRA, FIA & major banks
- ✓ Recognises JazzCash & Easypaisa patterns
- ✓ English, Urdu & Roman-Urdu evidence patterns

THE PEOPLE

Meet the team



Showkat Raja

Team Lead



Moneka Meghwar

Documentation Lead



Syed Mehtab Ali

Presentation Lead



Ayesha Ashraf

Research Work



Syed Alam Shah Bukhari

Research Work



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